

Student Name: _____

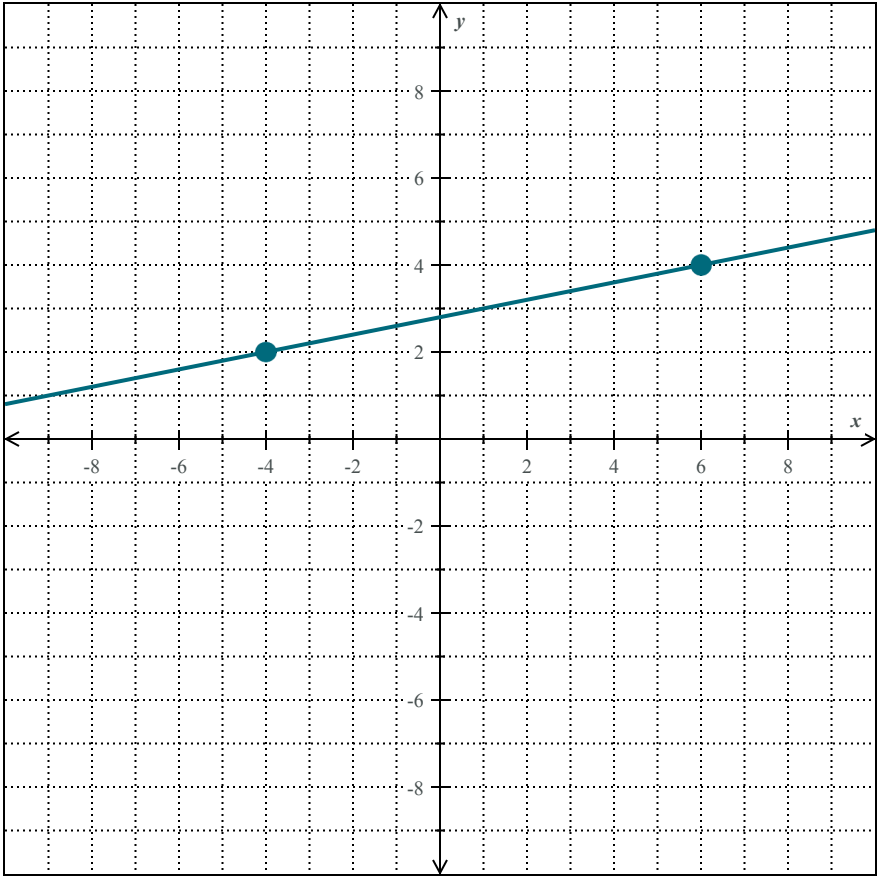
Class Name: **College Algebra FA26 - 014 MW 2:30**

Number of Questions: **10**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 10

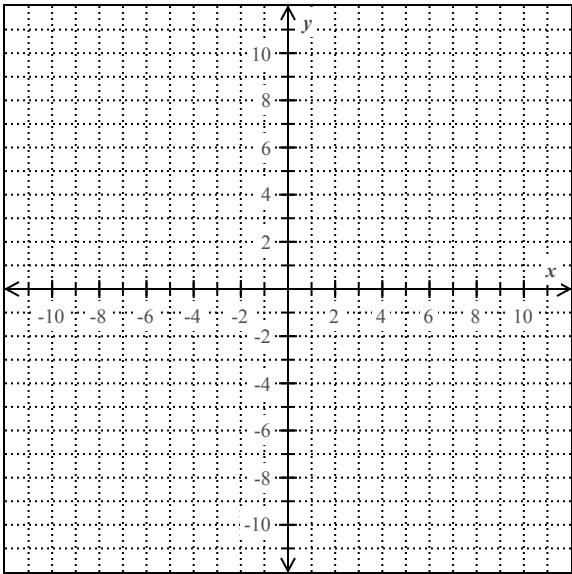
Find an equation for the line below.



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Graph the equation.

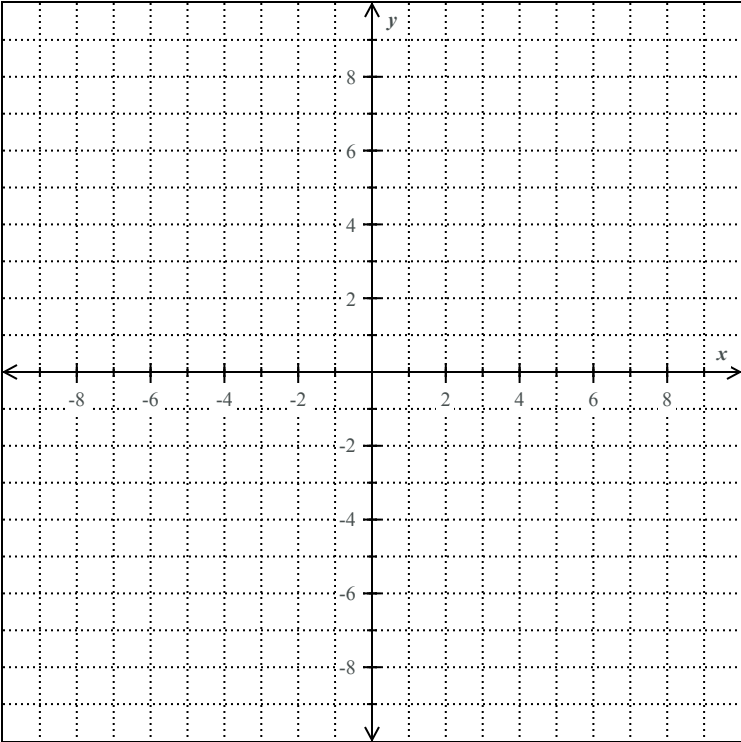
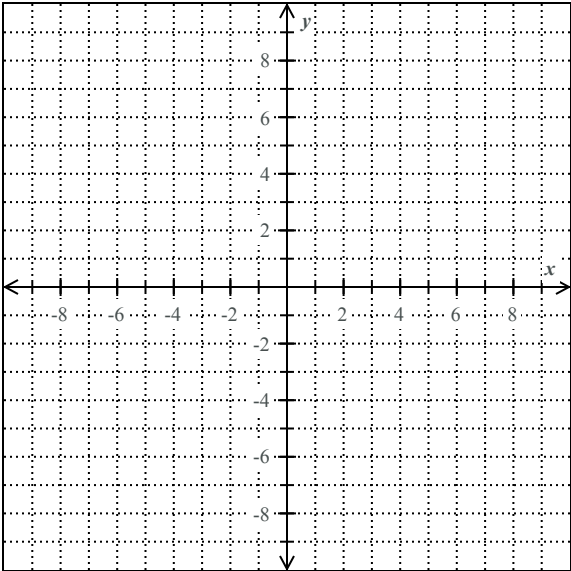
$$y=2|x|$$



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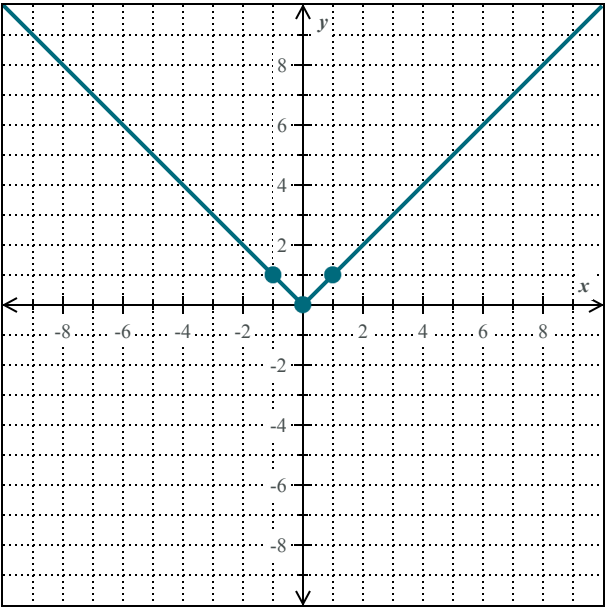
Graph the parabola.

$y = (x + 1)^2 + 2$



Question 4 of 10

Below is the graph of $y = |x|$. Translate it to make it the graph of $y = |x + 3| - 4$.



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The function h is defined as follows.

$$h(x) = x^2 + 3$$

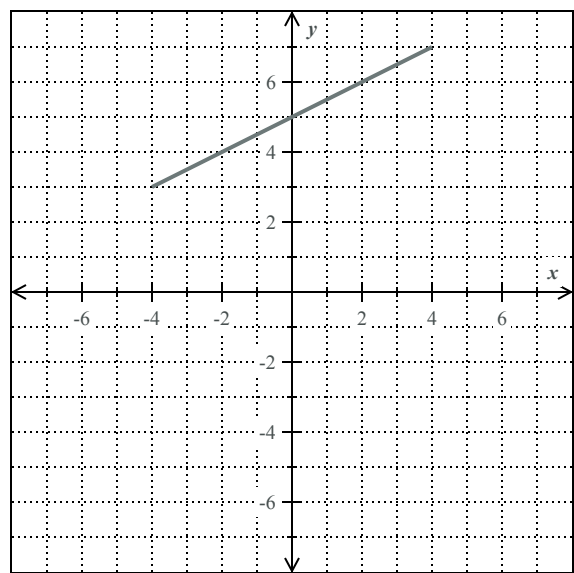
If the graph of h is translated vertically downward by 3 units, it becomes the graph of a function g .

Find the expression for $g(x)$.

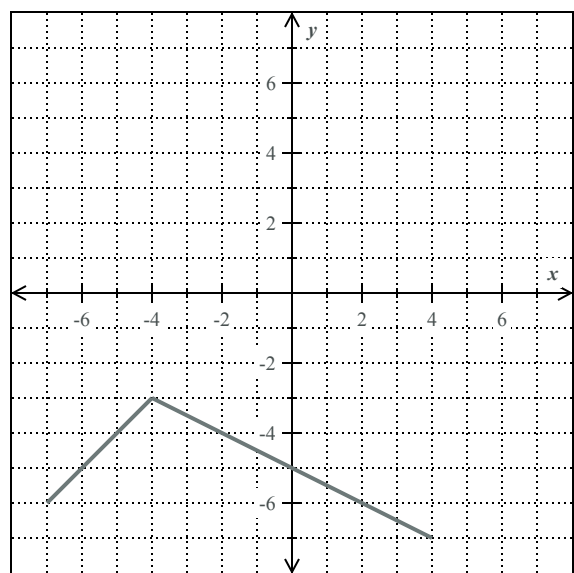
$g(x) =$

Question 6 of 10

(a) The graph of $y=f(x)$ is shown. Draw the graph of $y=f(-x)$.



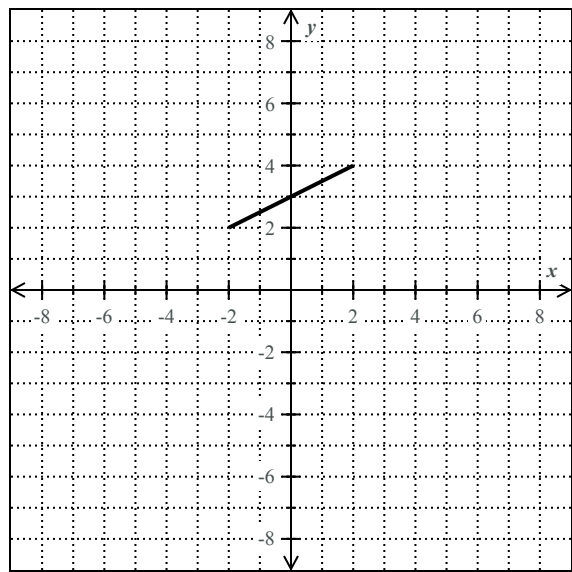
(b) The graph of $y=g(x)$ is shown. Draw the graph of $y=-g(x)$.



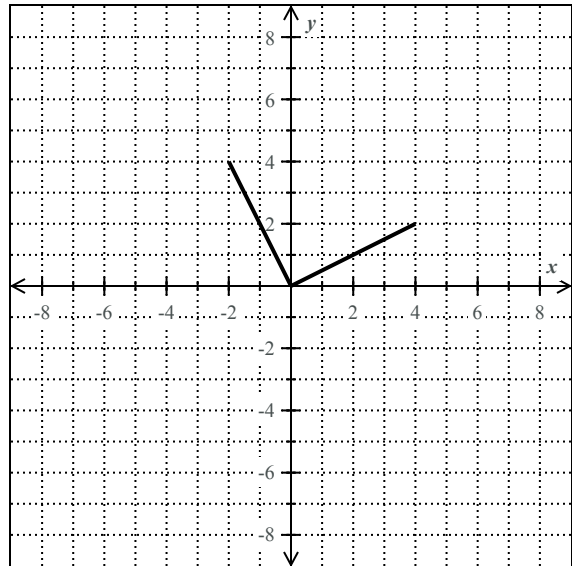
Question 7 of 10

Transform each graph as specified below.

(a) The graph of $y=f(x)$ is shown. Draw the graph of $y=f\left(\frac{1}{2}x\right)$.



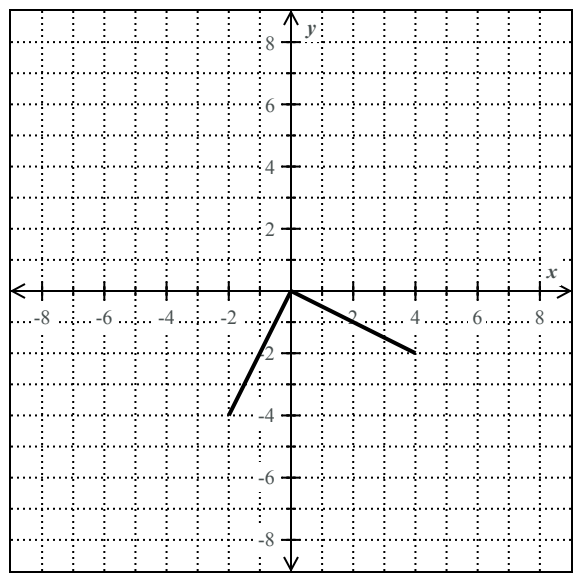
(b) The graph of $y=g(x)$ is shown. Draw the graph of $y=2g(x)$.



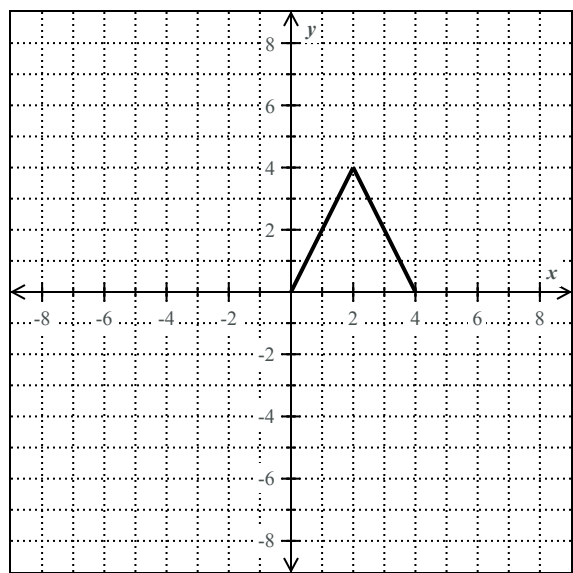
Question 8 of 10

Complete the following.

(a) The graph of $y = g(x)$ is shown. Draw the graph of $y = 2g(x) + 4$.



(b) The graph of $y = h(x)$ is shown. Draw the graph of $y = -h(x - 3)$.



Question 9 of 10

Suppose that the function g is defined on the interval $[-2, 2)$ as follows.

$$g(x) = \begin{cases} -2 & \text{if } -2 \leq x < -1 \\ -1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x < 2 \end{cases}$$

Find $g(-2)$, $g(0.75)$, and $g(1)$.

$$g(-2) = \underline{\hspace{2cm}}$$

$$g(0.75) = \underline{\hspace{2cm}}$$

$$g(1) = \underline{\hspace{2cm}}$$

Question 10 of 10

Three functions are given below.

For each, decide whether it is an even function, an odd function, or neither.

$f(x) = \sqrt[3]{9x^2}$ ☐ Even ☐ Odd ☐ Neither	$g(x) = 3x x $ ☐ Even ☐ Odd ☐ Neither	$h(x) = \frac{1}{x^2 + 7}$ ☐ Even ☐ Odd ☐ Neither
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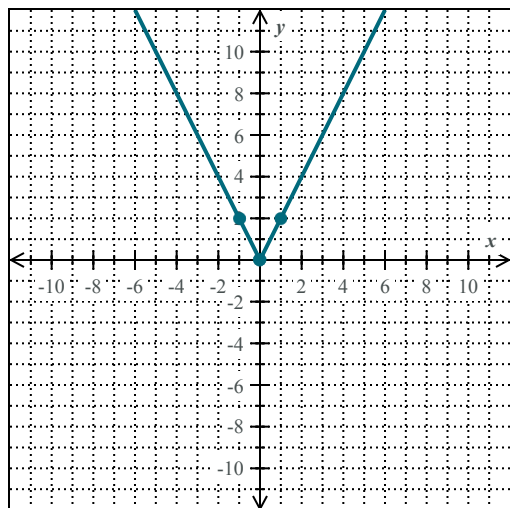
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Question 1 of 10

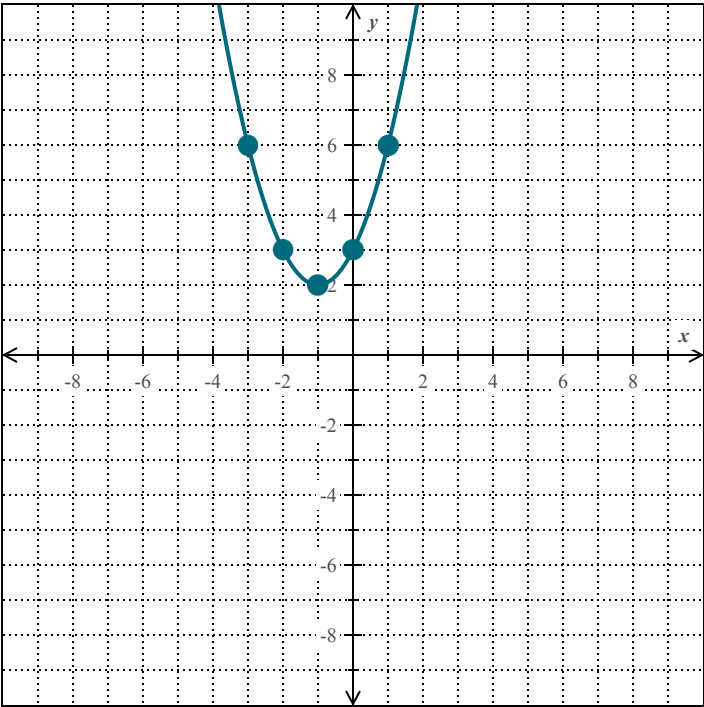
$$y = \frac{1}{5}x + \frac{14}{5}$$

Note: Other correct answers include $y - 2 = \frac{1}{5}(x + 4)$ and $x - 5y = -14$.

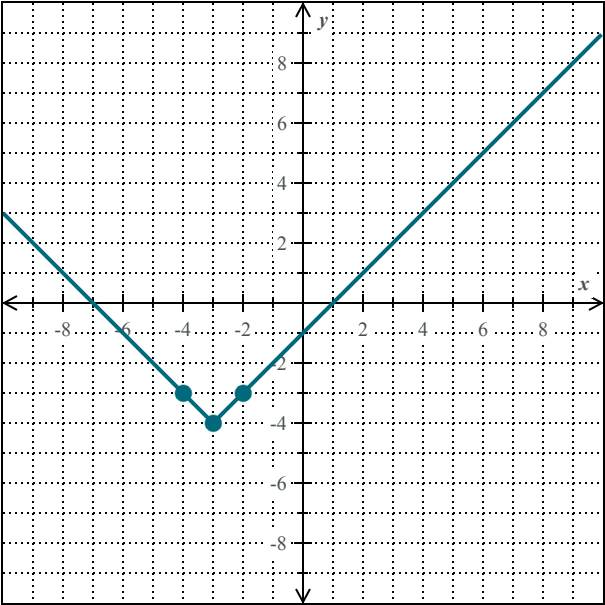
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Question 4 of 10

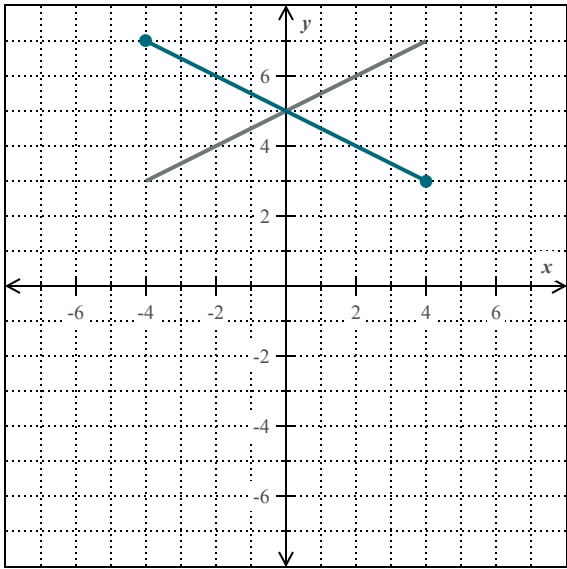


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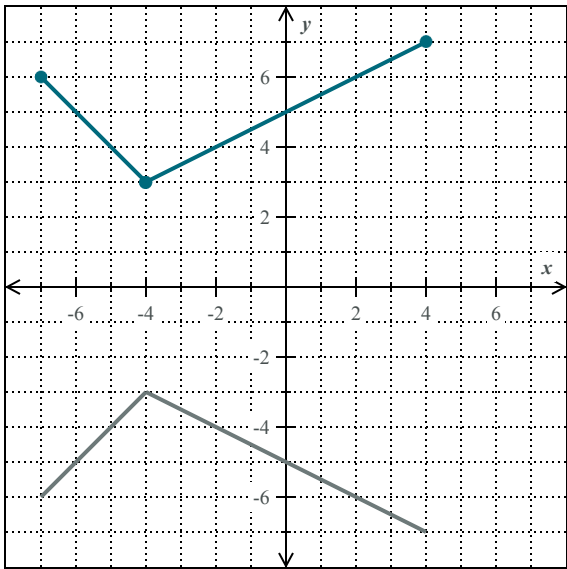
$g(x) = x^2$

Question 6 of 10

(a) The graph of $y=f(x)$ is shown. Draw the graph of $y=f(-x)$.

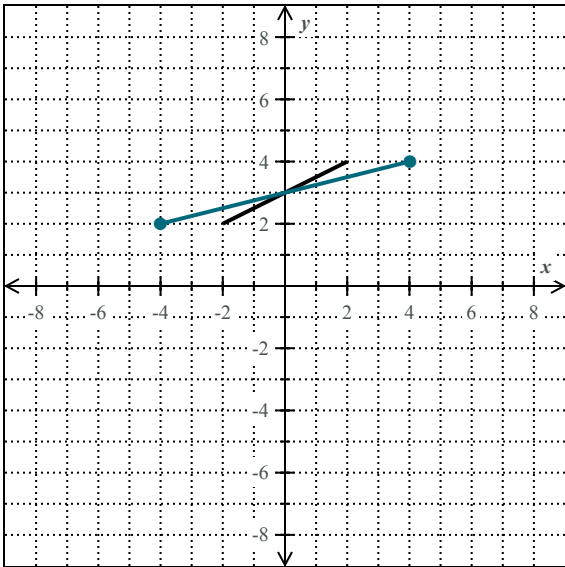


(b) The graph of $y=g(x)$ is shown. Draw the graph of $y=-g(x)$.

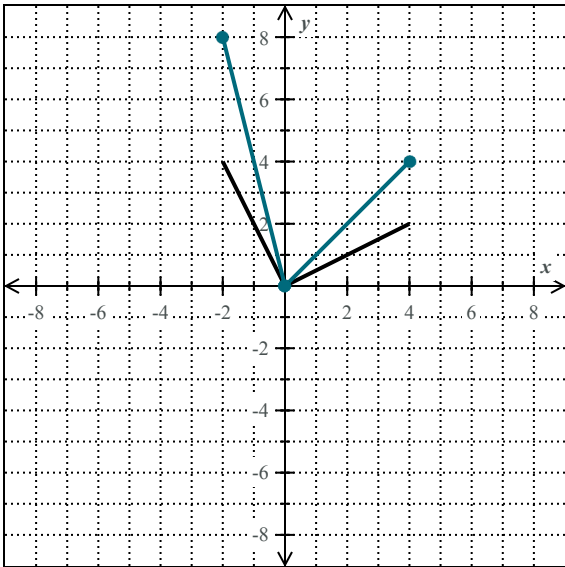


Question 7 of 10

(a) The graph of $y=f(x)$ is shown. Draw the graph of $y=f\left(\frac{1}{2}x\right)$.

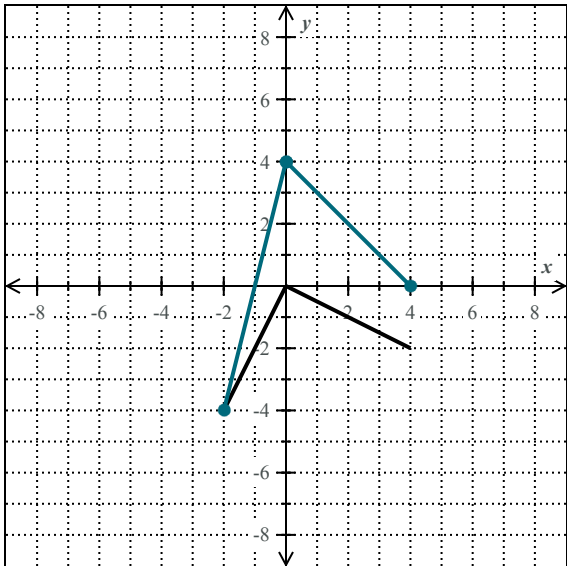


(b) The graph of $y=g(x)$ is shown. Draw the graph of $y=2g(x)$.

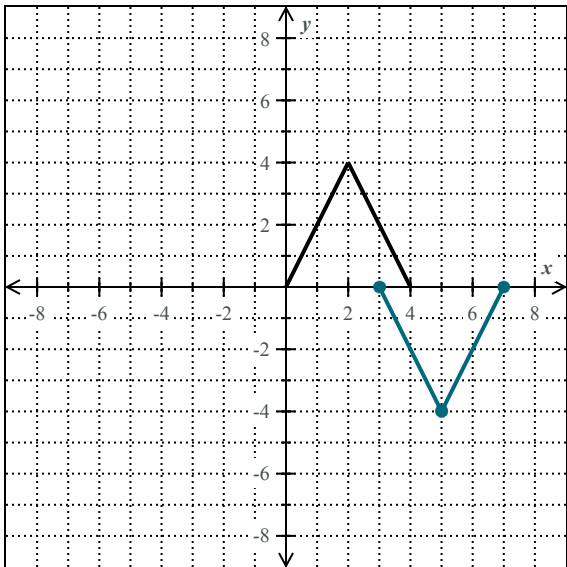


Question 8 of 10

(a) The graph of $y = g(x)$ is shown. Draw the graph of $y = 2g(x) + 4$.



(b) The graph of $y = h(x)$ is shown. Draw the graph of $y = -h(x - 3)$.



Question 9 of 10

$$g(-2) = -2$$
$$g(0.75) = 0$$
$$g(1) = 1$$

Question 10 of 10

$f(x) = \sqrt[3]{9x^2}$ ☒ Even ☐ Odd ☐ Neither	$g(x) = 3x x $ ☐ Even ☒ Odd ☐ Neither	$h(x) = \frac{1}{x^2 + 7}$ ☒ Even ☐ Odd ☐ Neither
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